



Q1. What are the differences between NE555 and LM555 ICs?

Mohd Waseem Ansari

A1. There is no difference between NE555 and LM555 ICs. Both these are 8-pin timer ICs with pin to pin compatibility. Both have the same sinking and sourcing output up to 200mA and maximum supply voltage of 18V. Both are available in PDIP as well as other packages. LM555CN is from National Semiconductor and SE555/NE555 from Signetics. Both work with similar arrangement of components at pins 2, 6 and 7 for monostable and astable modes. In short, they are interchangeable.

Q2. We want to set up a mini radio station for our institution and the local area within Bengaluru. It should have a range of about 16km but our budget for expenditure is ₹ 5000 only.

Praveen and Shubham

A2. Permission for community FM radio stations is granted by Ministry of Information & Broadcasting (Community Radio Station Cell) of the government of India. The licence is granted to well-established educational institutions including IITs/IIMs and non-profit organisations like civil society and voluntary organisations. Application forms and details of documents required can be downloaded from the website of Community Radio Stations Management Information System, Ministry of Information and Broadcasting.

Consultancy services are also available to provide entire range of services required for setting up a community radio station (CRC) starting with help in filing the application for licence and frequency allocation, up to the completion of the project of setting up the station. However, a budget of ₹ 5000 will be grossly insufficient for starting

a CRC as it will need studio facilities for proper functioning besides the transmitter.

However, if you are looking for just an FM or AM transmitter circuit, you will find quite a few in previous issues of EFY magazine.

Q3. I want to know what is the standard PCB track width and gap between tracks for normal and power supply lines? Which is the best free PCB design software available?

Vijay Kakul

A3. Proper distance between PCB tracks is critical to avoid flashover between electrical conductors. There are industry, mil and safety standards that prescribe different spacing and trace-width requirements depending on the voltage, current density, application and other factors. Standards for PCB industry developed by Institute for Interconnecting and Packaging Electronic Circuits (IPC), a US based trade association, are referred to world over. Note that these standards are voluntary rather than mandatory.

There are two factors to be considered: spacing between adjacent conductors and widths of the tracks. The conductor spacing considerations are generally based on breakdown voltage or flashover voltage between adjacent conductors. The conductor spacing is determined by the maximum voltage difference permissible between adjacent conductors, capacitive coupling and use of coatings.

In most of our applications, we make use of semiconductor devices operating at up to 24V normally. Therefore the problem of breakdown due to narrow spacing between conductors almost does not exist. Generally, a minimum gap of 0.18mm (0.007-inch) between tracks is enough, though 0.25mm (0.01-inch) is better. IPC recommended clearances adjacent to conductors for different voltages are given in Table I.

The PCB conductor thickness and width determines the current carry-

TABLE I
IPC-Recommended Clearances

Voltage	Coated board	Uncoated up to 10,000 ft	Uncoated over 10,000 ft
0-50	0.13mm	0.64mm	0.6mm
51-100	0.13mm	0.64mm	1.50mm
101-150	0.40mm	0.64mm	3.18mm
151-250	0.40mm	1.27mm	3.18mm

ing capacity of the track. The IPC standard for the conductor thickness and width of the common 1 oz/square-foot PCB (1oz/sq foot copper – 0.035mm thickness) for different current densities are

TABLE II
Conductor Thickness and Width

Current in A	Track width in mil/mm
1	10/0.25
2	30/0.76
3	50/1.27
4	80/2.03
5	110/2.79

given in the Table II. However, it is always advisable to use a bigger value due to the tolerance and variation of the PCB processes. If higher current carrying capacity is required, a 2 oz/square-foot or 3 oz/square-foot type of PCB is preferred. You can resort to thick soldering of tracks to increase the current carrying capacity of the track.

There are quite a few free software available for PCB designing, most popular of which are gEDA and EAGLE (Easily Applicable Graphical Layout Editor). Free version of EAGLE has, however, a limitation of size of PCB. You can make PCB layout up to the size of a post-card. It is very easy to use, with huge library of footprints of components. Although gEDA is open source software based on Linux with no restriction on size of PCB, it is not as widely used as EAGLE and the documentation tends to be scattered around and poorly organised in online community.

Answers compiled by EFY joint director (training), Col. N.C. Pande (Retd). Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender